Projection onto the l2 ball: Real Stability

Research Architecture Runtime

`operator/projection-l2-ball`

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Abstract

If a projector fixes feasible points and the closed ε-ball about x lies in the feasible set, then every x' in the ball is fixed. This is an identity certificate, not full Euclidean projection nonexpansiveness. Lean `LEAN_FULL`.

This theorem is: `Reduction`

1 Problem

`projection-l2-ball` is treated via a feasibility indicator / projection that fixes InSet points.

2 Stability notion

Interior feasible-ball identity under $|x' - x| \leq \varepsilon$.

3 Definitions

Assumptions: $\mathrm{proj}(z) = z$ whenever $\mathrm{InSet}(z)$, and $|y - x| \leq \varepsilon \Rightarrow \mathrm{InSet}(y)$.

4 Theorem

Theorem 1. *Under those assumptions, every $|x' - x| \leq \varepsilon$ satisfies $\mathrm{proj}(x') = x'$ and $\mathrm{proj}(x) = x$. Sharpness: if some ball point is infeasible, the ball-feasibility hypothesis fails.*

5 Intuition

Deep inside the feasible set, projection is the identity, so small moves stay fixed points.

6 Examples

Example 2. If InSet $= [-1, 1]$, $x = 0$, $\varepsilon = 0.25$, every x' in the ball is feasible and fixed by identity projection.

7 Proof

Take $|x' - x| \le \varepsilon$. Feasibility of the ball gives $\mathrm{InSet}(x')$ and $\mathrm{InSet}(x)$. Fixing feasible points yields $\mathrm{proj}(x') = x'$ and $\mathrm{proj}(x) = x$ (Theorem 1). Sharpness is the contrapositive of ball feasibility. **Limitation:** this does not prove general Euclidean projection nonexpansiveness.

8 Formal statement

Lean module
`Research.Operators.ProjectionL2Ball.Preservation`

Source file
`lean/Research/Operators/ProjectionL2Ball/Preservation.lean`

Theorems

- `projection\_l2\_ball\_feasible\_ball\_identity`
- `projection\_l2\_ball\_feasible\_ball\_sharpness`

Note
Uses Projection.FeasibleId (not Euclidean nonexpansiveness).

Certificate
`lean/certificates/projection-l2-ball/projection-l2-ball-feasible-ball-identity`

Status
`LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`)

9 Proof dependencies

- `Projection.FeasibleId`.

- No Euclidean nonexpansiveness claim.

10 Consequences

Honest limited certificate for simplex/ℓ_p-ball style operators in this library. Related: projection-interval, coordinate-clipping.